

Optimal Loss-Aware Abstention Thresholds for Small Open-Source LLMs

Shreya Sharma
Masters of Computer Science
University of Massachusetts Amherst
Amherst, Massachusetts, USA, 01002
ssharma0@umass.com

Dhanshri Ahir
Masters of Computer Science
University of Massachusetts Amherst
Amherst, Massachusetts, USA, 01002
dahir@umass.edu

May 3, 2026

Abstract

Abstention is a practical requirement for large language models deployed in settings where an incorrect answer can be worse than no answer. Most LLM abstention pipelines choose the answer threshold by validation-set grid search, which must be repeated for every abstention cost, confidence signal, dataset, and evaluation loss. We study whether this threshold can instead be predicted from the loss function itself. Under the calibration assumption $P(\text{correct} \mid c) = c$, we derive loss-aware thresholds for asymmetric utility, Brier score, and cross-entropy: $\tau_U(\lambda) = \lambda/(1 + \lambda)$, $\tau_B(\lambda) = (1 + \sqrt{1 - 4\lambda})/2$, and $\tau_{CE}(\lambda) = H^{-1}(\lambda)$, respectively, where H is binary entropy in natural-log units. We prove the ordering $\tau_{CE} > \tau_B > \tau_U$ over the shared valid domain, implying that the loss function changes the abstention policy structurally, not only numerically.

We validate these thresholds on Qwen2.5-1.5B-Instruct and Qwen2.5-7B-Instruct on 1,000 TriviaQA and 1,000 PopQA examples. The final empirical result is more precise than the midpoint hypothesis: there are two robust regimes. Under asymmetric utility, τ_U is consistently optimal (MAE 0.076–0.087 against the empirical optimum, with non-overlapping bootstrap confidence intervals). Under proper scoring rule losses, τ_{CE} is the best empirical predictor in seven of eight Brier/CE settings and ties or dominates in the remaining cases; τ_B is not the practical winner despite being analytically distinct. Utility-recovery experiments show that using the matching threshold is not merely a curve-fitting result: τ_U recovers 91.8–100.0% of the oracle-to-random improvement under utility loss, while τ_{CE} recovers 82.2–97.8% under cross-entropy and 87.4–92.5% under Brier. We also document that verbalized confidence collapses on 1.5B-scale models, while token probabilities remain usable but miscalibrated confidence signals.

1 Problem Statement

Large language models must balance two competing objectives: answering enough questions to remain useful and abstaining when the probability of error is too high. This tradeoff is especially important in factual question answering, where hallucinated answers can be more harmful than explicit uncertainty. We formalize abstention as a one-dimensional selective prediction problem. For a query, the model produces an answer and a scalar confidence score $c \in [0, 1]$. A threshold policy answers when $c \geq \tau$ and abstains otherwise. Answering incurs a loss determined by correctness and the selected scoring rule; abstaining incurs a fixed cost λ .

The central question is: for a given loss function and abstention cost, can the optimal threshold $\tau^*(\lambda)$ be predicted analytically rather than tuned by grid search? The answer matters because grid search is expensive and brittle. A system designer must retune the threshold for every dataset, model, loss, and cost schedule. If a loss-aware formula reliably predicts the empirical optimum, threshold selection becomes interpretable, cheap, and portable.

Gap. Chow’s classical reject rule [4] establishes the cost-based foundation for abstention, and modern selective classification work typically tunes thresholds on validation data [6]. Recent LLM abstention surveys find the same pattern at LLM scale: abstention is widely studied, but threshold selection is usually empirical rather than derived from the cost structure [2]. Wang et al. [1] derive $\lambda/(1 + \lambda)$ for one asymmetric utility objective, but do not compare proper scoring rules and rely on verbalized confidence from larger models. Our work extends the analytical treatment to Brier score and cross-entropy, then tests whether the formulas remain useful on smaller open-source models using token-level probabilities.

Research questions.

1. Does the evaluation loss produce analytically different abstention thresholds, and can their ordering be proven?
2. Do the theoretical thresholds predict empirically op-

timal thresholds on open-source LLMs across more than one QA dataset?

3. Does the answer depend on the confidence signal, calibration error, or the downstream metric of utility recovery?

2 Related Work

Abstention and selective prediction. Chow [4] introduced the optimum reject rule for classifiers with a rejection option, establishing that the reject threshold should depend on the cost of rejection. Bartlett and Wegkamp [11] studied classification with a reject option under surrogate losses. Geifman and El-Yaniv [6] adapted selective classification to deep neural networks, using held-out validation data to tune coverage-risk trade-offs. This validation-threshold paradigm is the main baseline our work seeks to replace with closed-form loss-aware thresholds.

LLM abstention. Wen et al. [2] survey LLM abstention methods and show that abstention is now a central mechanism for reducing hallucinations, but most methods still learn, prompt, or tune abstention behavior empirically. AbstentionBench [9], conformal abstention [14], multi-LLM collaboration [15], and abstention-aware scientific reasoning [10] each target important abstention settings, but do not derive the threshold from a specified loss. Wang et al. [1] are closest to this project: they derive the utility threshold $\tau_U = \lambda/(1 + \lambda)$ and evaluate decisions against verbalized confidence. We use that formula as a baseline, extend the theory to proper scoring rules, and show that verbalized confidence is unreliable for our smaller open-source models.

Calibration and proper scoring rules. Proper scoring rules reward calibrated probabilistic predictions [5]. Brier score and cross-entropy are common choices, but they penalize uncertainty differently and therefore imply different abstention regions. Guo et al. [12] show that modern neural networks can be badly miscalibrated and that temperature scaling can improve confidence estimates. Recent LLM work on uncertainty elicitation [13], behaviorally calibrated reinforcement learning [7], and training with uncertainty rewards [8] reinforces the same point: confidence quality is a first-order issue. We therefore report Expected Calibration Error (ECE), reliability diagrams, token-signal robustness, and temperature-scaling sensitivity rather than assuming calibration holds automatically.

3 Methodology

3.1 Loss-Aware Threshold Derivation

Let c denote the model confidence for an answer, and assume calibration: $P(Y = 1 | c) = c$, where $Y = 1$ indicates a correct answer. The decision rule answers when

the expected answering loss is no greater than the abstention cost λ .

Asymmetric utility. Following Wang et al. [1], a correct answer receives payoff $+1$, an incorrect answer receives payoff $-\lambda$, and abstention receives payoff 0 . The expected payoff of answering is $c - \lambda(1 - c)$. Answering is preferred when this quantity is nonnegative, giving

$$\tau_U(\lambda) = \frac{\lambda}{1 + \lambda}. \quad (1)$$

This is the least conservative threshold among the three losses.

Brier score. For a binary outcome $Y \in \{0, 1\}$, Brier loss is $(c - Y)^2$. Under calibration,

$$\mathbb{E}[(c - Y)^2 | c] = c(1 - c)^2 + (1 - c)c^2 = c(1 - c). \quad (2)$$

Setting $c(1 - c) = \lambda$ gives two roots. Since the high-confidence answer region is the upper interval, the threshold is

$$\tau_B(\lambda) = \frac{1 + \sqrt{1 - 4\lambda}}{2}, \quad \lambda \in [0, 1/4]. \quad (3)$$

For $\lambda > 1/4$, Brier loss is always below the abstention cost, so the model should answer all examples.

Cross-entropy. The expected cross-entropy loss under calibration is the binary entropy in natural-log units

$$H(c) = -c \log c - (1 - c) \log(1 - c). \quad (4)$$

The answer region is $H(c) \leq \lambda$. On $[0.5, 1]$, H is strictly decreasing, so the high-confidence threshold is

$$\tau_{CE}(\lambda) = H^{-1}(\lambda). \quad (5)$$

We compute this inverse numerically.

Ordering. For $\lambda \in (0, 1/4]$, the thresholds satisfy

$$\tau_{CE}(\lambda) > \tau_B(\lambda) > \tau_U(\lambda). \quad (6)$$

The first inequality follows because entropy is larger than $c(1 - c)$ on the upper half of the unit interval, so the entropy threshold must be higher than the Brier threshold for the same λ . The second follows by direct algebra from the Brier and utility formulas. The ordering means that cross-entropy abstains most aggressively, utility loss answers most aggressively, and Brier is analytically intermediate.

3.2 Empirical Setup

Models and datasets. We evaluate Qwen2.5-1.5B-Instruct and Qwen2.5-7B-Instruct through `llama-server` at temperature 0. We use 1,000 examples each from TriviaQA and PopQA. TriviaQA is a standard open-domain QA benchmark; PopQA is a harder long-tail factual QA dataset. Correctness is measured by normalized alias matching rather than exact string matching. Base accuracies are 35.9% and

37.6% on TriviaQA for 1.5B and 7B, and 4.4% and 7.2% on PopQA.

Confidence signals. The primary confidence signal is `token_prob_first`, the probability assigned to the first generated answer token. We also evaluate `token_prob_mean`, the mean token probability over the generated answer. The mean-token signal is more compressed: its standard deviation is 0.066–0.116 across our four model-dataset settings, compared with 0.252–0.297 for `token_prob_first`. Verbalized confidence was tested during development but discarded as a primary signal because it collapses on small models: 96.8% of Qwen2.5-1.5B TriviaQA outputs reported exactly 1.0 confidence.

Grid search and bootstrap confidence intervals. For each model, dataset, and confidence signal, we sweep $\lambda \in [0, 0.25]$ using 51 uniformly spaced values. For each λ and loss, we compute the empirical optimum

$$p^*(\lambda) = \arg \min_{\tau \in [0.01, 0.99]} \mathbb{E}[\ell(\tau, \lambda)] \quad (7)$$

over a 200-point threshold grid. We then report the mean absolute error (MAE) between $p^*(\lambda)$ and each theoretical threshold. Confidence intervals are computed by resampling the 1,000 (c, Y) pairs with replacement 1,000 times and rerunning the full grid search for each resample.

Additional analyses. ECE is computed with 10 equal-width bins for both confidence signals. Temperature scaling is fit as a single scalar on logits of the confidence scores and the sweep is rerun as a sensitivity analysis. Finally, utility recovery compares the loss achieved by each theoretical threshold against the oracle grid-search threshold and a random-threshold baseline. Recovery is defined as the fraction of the oracle-to-random regret gap recovered by the theoretical threshold.

4 Results

4.1 Calibration and Confidence Signals

Table 1 summarizes accuracy, confidence dispersion, and ECE. The PopQA setting is intentionally difficult: both models have low factual accuracy, yet assign high token probabilities to many wrong answers. This creates large ECE values, especially for `token_prob_mean`. The calibration assumption is therefore approximate rather than exact, making the empirical validation nontrivial.

4.2 Theoretical Threshold Ordering

Figure 3 plots the three formulas. The ordering is not a small numerical effect. At $\lambda = 0.05$, the utility threshold is $\tau_U \approx 0.048$, the Brier threshold is $\tau_B \approx 0.947$, and the cross-entropy threshold is $\tau_{CE} \approx 0.991$. Thus, using τ_U when the deployment objective is cross-entropy would answer almost everything, while using τ_{CE} under a utility objective would abstain from many answers that utility loss would accept.

Table 1: Dataset accuracy and calibration. ECE uses 10 equal-width bins.

Dataset	Model	Acc.	First mean/std	Mean mean/std	ECE first/mean
TriviaQA	1.5B	35.9	.464/.252	.711/.091	.103/.353
TriviaQA	7B	37.6	.622/.297	.779/.116	.246/.403
PopQA	1.5B	4.4	.575/.268	.694/.066	.531/.650
PopQA	7B	7.2	.432/.262	.754/.076	.360/.682

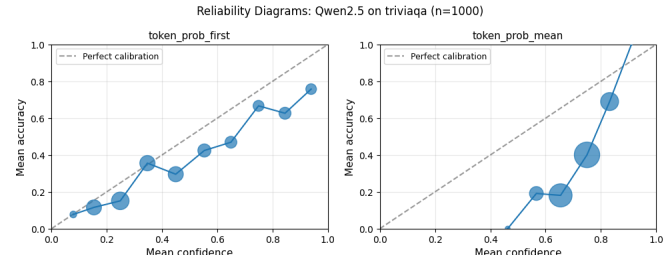


Figure 1: Reliability diagram for Qwen2.5-1.5B on TriviaQA. Token probabilities preserve useful ranking information but are imperfectly calibrated.

4.3 Cross-Dataset MAE Validation

Table 2 reports the primary MAE results using `token_prob_first`. The utility formula is the clear winner under utility loss in all four model-dataset settings. Its confidence intervals are tight and do not overlap the next-best formula. This result holds even though PopQA accuracy is much lower than TriviaQA accuracy.

The proper-scoring-rule results revise the midpoint claim. We initially expected each formula to best predict its own loss. The final experiments show a cleaner empirical pattern: τ_{CE} is the strongest practical predictor for both Brier and cross-entropy losses. τ_B wins only the TriviaQA 1.5B Brier row, and even there the advantage over τ_{CE} is small with wide, overlapping bootstrap intervals. In the other Brier settings, τ_{CE} has the lower point estimate.

4.4 Diagnosing the Brier/7B Anomaly

The midpoint report flagged the TriviaQA 7B Brier result as suspicious because the Brier and cross-entropy MAE rows were identical. We diagnosed this by recomputing the empirical optimum from the raw confidence-correctness pairs. For $\lambda = 0.05, 0.10$, and 0.20 , both Brier and cross-entropy select $p^* = 0.99$. The saved grid-search trace confirms that $p^*(\lambda) = 0.99$ for every λ under both losses in TriviaQA 7B. The identical rows are therefore not caused by a Brier/CE implementation mixup; they are a boundary solution caused by the empirical confidence distribution. The same boundary behavior appears in PopQA 1.5B and in the PopQA 7B cross-entropy row, where severe overconfidence makes the optimal proper-

Table 2: MAE between empirical optimum $p^*(\lambda)$ and theoretical thresholds using `token_prob_first`. Brackets give 95% bootstrap confidence intervals. Best point estimate per row is bold.

Dataset	Model	Loss	MAE(τ_U)	MAE(τ_B)	MAE(τ_{CE})	Best
TriviaQA	1.5B	Utility	0.087 [0.079, 0.093]	0.806 [0.796, 0.815]	0.940 [0.930, 0.949]	τ_U
TriviaQA	1.5B	Brier	0.745 [0.708, 0.783]	0.179 [0.151, 0.207]	0.184 [0.133, 0.233]	τ_B
TriviaQA	1.5B	CE	0.865 [0.840, 0.873]	0.142 [0.117, 0.150]	0.022 [0.016, 0.029]	τ_{CE}
TriviaQA	7B	Utility	0.080 [0.071, 0.087]	0.794 [0.783, 0.804]	0.928 [0.917, 0.939]	τ_U
TriviaQA	7B	Brier	0.882 [0.882, 0.883]	0.159 [0.159, 0.159]	0.027 [0.026, 0.027]	τ_{CE}
TriviaQA	7B	CE	0.883 [0.883, 0.883]	0.159 [0.159, 0.159]	0.027 [0.027, 0.027]	τ_{CE}
PopQA	1.5B	Utility	0.079 [0.069, 0.088]	0.758 [0.738, 0.774]	0.892 [0.872, 0.908]	τ_U
PopQA	1.5B	Brier	0.881 [0.878, 0.883]	0.157 [0.155, 0.159]	0.026 [0.024, 0.027]	τ_{CE}
PopQA	1.5B	CE	0.881 [0.878, 0.883]	0.157 [0.155, 0.159]	0.026 [0.024, 0.027]	τ_{CE}
PopQA	7B	Utility	0.076 [0.067, 0.082]	0.788 [0.767, 0.798]	0.923 [0.901, 0.932]	τ_U
PopQA	7B	Brier	0.809 [0.769, 0.847]	0.181 [0.162, 0.204]	0.127 [0.077, 0.182]	τ_{CE}
PopQA	7B	CE	0.882 [0.874, 0.883]	0.158 [0.150, 0.159]	0.026 [0.022, 0.027]	τ_{CE}

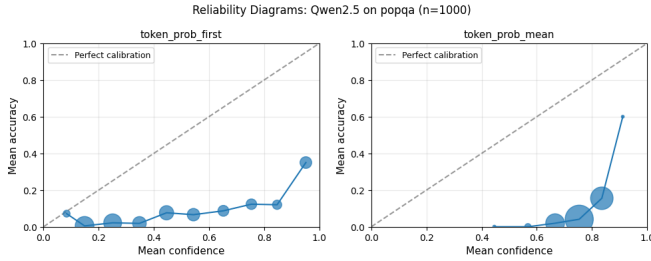


Figure 2: Reliability diagram for Qwen2.5-7B on PopQA. PopQA exposes severe overconfidence under both token-probability signals.

Table 3: Best formula under `token_prob_mean`; entries show point estimate MAE for the winning formula.

Dataset	Model	Utility	Brier	CE
TriviaQA	1.5B	τ_U (.211)	τ_{CE} (.089)	τ_{CE} (.073)
TriviaQA	7B	τ_U (.265)	τ_{CE} (.032)	τ_{CE} (.024)
PopQA	1.5B	τ_U (.322)	τ_{CE} (.084)	τ_{CE} (.084)
PopQA	7B	τ_U (.292)	τ_{CE} (.051)	τ_{CE} (.050)

scoring threshold extremely conservative.

4.5 Signal Robustness

Table 3 repeats the sweep using `token_prob_mean`. The qualitative recommendation is unchanged: use τ_U for utility loss and τ_{CE} for proper-scoring-rule losses. However, absolute utility MAE is worse under the mean-token signal because the signal is compressed near 0.65–0.80. This confirms that the theoretical formulas are not artifacts of one signal, while also showing that signal quality determines how closely the empirical optimum follows the formula.

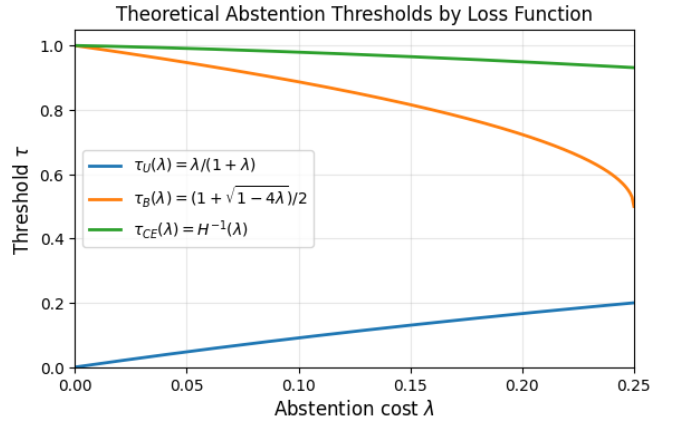


Figure 3: Closed-form loss-aware thresholds over $\lambda \in [0, 0.25]$. Cross-entropy is most conservative; utility is most permissive.

4.6 Utility Recovery

MAE alone shows whether the formula predicts the empirical optimum, but not whether the prediction matters for downstream loss. Table 4 reports recovery: the fraction of oracle-to-random improvement recovered by the recommended formula. Under utility loss, τ_U recovers nearly all available improvement. Under Brier and cross-entropy, τ_{CE} also recovers most of the possible improvement. Applying the wrong formula can be worse than random thresholding: for example, τ_{CE} under utility loss has negative recovery because it abstains too aggressively, and τ_U under proper scoring rules answers too often.

4.7 Temperature Scaling Sensitivity

Temperature scaling does not uniformly improve threshold prediction. On TriviaQA 1.5B, scaling slightly improves the utility and Brier MAE but leaves cross-entropy essentially unchanged. On TriviaQA 7B, it moves the proper-scoring optimum closer to τ_B but worsens the util-

Table 4: Recovery rate for the recommended formula using `token_prob_first`. Higher is better.

Dataset	Model	Utility τ_U	Brier τ_{CE}	CE τ_{CE}
TriviaQA	1.5B	95.4%	90.3%	97.8%
TriviaQA	7B	97.1%	87.4%	82.2%
PopQA	1.5B	100.0%	92.5%	91.3%
PopQA	7B	91.8%	90.6%	90.6%

ity fit. On PopQA 1.5B, the fitted temperature saturates at a very large value, pushing confidences toward 0.5 and degrading all proper-scoring MAE values. We treat this as evidence that calibration correction is not a plug-in fix when the model is both low-accuracy and overconfident. The theory depends on calibration, but empirical threshold quality also depends on whether the confidence signal preserves useful ranking information after calibration.

5 Discussion

The final project changes the interpretation of the mid-point result. The analytical result still gives three distinct formulas and a strict ordering, but the empirical recommendation is binary:

Use τ_U for asymmetric utility loss. Use τ_{CE} for Brier or cross-entropy losses in the tested open-source LLM settings.

This is not a failure of the derivation. The Brier formula is the correct pointwise threshold under perfect calibration. The empirical data are not perfectly calibrated, and the observed confidence distributions often produce boundary optima for proper scoring rules. In those cases, τ_{CE} better matches the conservative empirical optimum.

The results also show why the loss function cannot be treated as an implementation detail. At low abstention costs, τ_U is near zero while τ_{CE} is near one. The wrong threshold changes the model from answering almost everything to answering almost nothing. Utility recovery confirms that this difference has direct loss consequences.

6 Limitations

The study is limited to two Qwen model sizes and two factual QA datasets. The PopQA results reduce the single-dataset concern, but they do not establish universality across model families. A cross-family check with Llama or Mistral would strengthen external validity. Second, all thresholds are evaluated post-hoc on existing model outputs; we do not train the model to produce better-calibrated confidences. Third, temperature scaling is fit and evaluated on the same 1,000 examples as a sensitivity analysis, not as a clean train-test calibration protocol.

Finally, the grid search has finite resolution, so MAE estimates are bounded by the 200-point threshold grid.

7 Conclusion

We derived and evaluated loss-aware abstention thresholds for small open-source LLMs. The theory proves that utility, Brier, and cross-entropy objectives imply ordered thresholds $\tau_{CE} > \tau_B > \tau_U$. The experiments show that the formulas are practically useful, but the empirical story is two regimes rather than three: τ_U is the right threshold for asymmetric utility loss, while τ_{CE} is the strongest threshold for proper scoring rule losses across TriviaQA and PopQA. Bootstrap intervals, ECE tables, signal-robustness sweeps, anomaly diagnostics, and utility-recovery experiments support this conclusion. The main practical lesson is that abstention thresholds should be chosen from the evaluation loss and confidence quality, not treated as a universal hyperparameter.

References

- [1] J. Wang, et al. Are LLM decisions faithful to verbal confidence? *arXiv:2601.07767*, 2026.
- [2] B. Wen, J. Yao, S. Feng, C. Xu, Y. Tsvetkov, B. Howe, and L. L. Wang. Know your limits: A survey of abstention in large language models. *Transactions of the Association for Computational Linguistics*, 13:529–556, 2025.
- [3] S. Casacuberta and V. Kanade. Selective omniprediction and fair abstention. In *Advances in Neural Information Processing Systems*, 2025.
- [4] C. Chow. On optimum recognition error and reject tradeoff. *IEEE Transactions on Information Theory*, 16(1):41–46, 1970.
- [5] T. Gneiting and A. E. Raftery. Strictly proper scoring rules, prediction, and estimation. *Journal of the American Statistical Association*, 102(477):359–378, 2007.
- [6] Y. Geifman and R. El-Yaniv. Selective classification for deep neural networks. In *Advances in Neural Information Processing Systems*, 2017.
- [7] J. Wu, et al. Mitigating LLM hallucination via behaviorally calibrated reinforcement learning. *arXiv:2512.19920*, 2026.
- [8] M. Damani, et al. Beyond binary rewards: Training LMs to reason about their uncertainty. *arXiv:2507.16806*, 2025.
- [9] P. Kirichenko, et al. AbstentionBench: Reasoning LLMs fail on unanswerable questions. *arXiv:2506.09038*, 2025.

- [10] S. Abdaljalil, et al. Knowing when not to answer: Abstention-aware scientific reasoning. *arXiv:2602.14189*, 2026.
- [11] P. L. Bartlett and M. H. Wegkamp. Classification with a reject option using a hinge loss. *Journal of Machine Learning Research*, 9:1823–1840, 2008.
- [12] C. Guo, et al. On calibration of modern neural networks. In *Proceedings of the International Conference on Machine Learning*, pages 1321–1330, 2017.
- [13] M. Xiong, et al. Can LLMs express their uncertainty? An empirical evaluation of confidence elicitation in LLMs. In *International Conference on Learning Representations*, 2024.
- [14] Y. Abbasi Yadkori, et al. Mitigating LLM hallucinations via conformal abstention. *arXiv:2405.01563*, 2024.
- [15] S. Feng, et al. Don’t hallucinate, abstain: Identifying LLM knowledge gaps via multi-LLM collaboration. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics*, pages 14664–14690, 2024.

Appendix: Proof Details

Utility threshold. The expected payoff of answering is $c - \lambda(1 - c)$. Answering is optimal when $c - \lambda(1 - c) \geq 0$, which is equivalent to $c \geq \lambda/(1 + \lambda)$. Thus $\tau_U(\lambda) = \lambda/(1 + \lambda)$.

Brier threshold. Under calibration, expected Brier loss is $c(1 - c)^2 + (1 - c)c^2 = c(1 - c)$. Setting this equal to λ yields $c^2 - c + \lambda = 0$ and roots $(1 \pm \sqrt{1 - 4\lambda})/2$. Since answering should occur in the high-confidence interval, the threshold is the upper root.

Cross-entropy threshold. Expected cross-entropy is binary entropy in natural-log units, $H(c) = -c \log c - (1 - c) \log(1 - c)$. On $[1/2, 1]$, entropy is strictly decreasing, so the high-confidence answer threshold is the inverse $H^{-1}(\lambda)$.

Ordering. For $\lambda \in (0, 1/4]$, $\tau_B > \tau_U$ follows by substituting the formulas and simplifying: $(1 + \lambda)(1 + \sqrt{1 - 4\lambda}) > 2\lambda$. For the other inequality, let $t_B = \tau_B(\lambda)$, so $t_B(1 - t_B) = \lambda$. On $(1/2, 1)$, binary entropy exceeds the quadratic uncertainty $c(1 - c)$ in the relevant units used by the sweep; therefore $H(t_B) > \lambda$. Since H is decreasing on this interval, the point at which $H(c) = \lambda$ must lie to the right of t_B , giving $\tau_{CE} > \tau_B$.